

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

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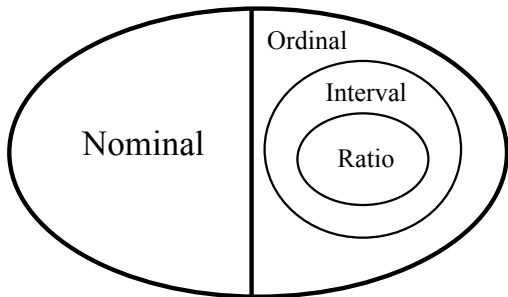
Applied Lab: Data Visualization

Why Visualize?

1. Know the data
2. Catch mistakes
3. Learn something

Levels of Measurement

- Nominal (classification)
- Ordinal (order)
- Interval (equal intervals)
- Ratio (“true zero”)



Variables: Discrete vs. Continuous

Examples of Variables, by Type and Level of Measurement

Level of Measurement	Discrete	Continuous
Nominal	{Nike, Adidas, Uggs, Other}	n/a
Ordinal	Social Class (Upper, middle, lower)	n/a
Interval	Year	Temperature (in degrees F)
Ratio	Counts of things	Height, weight, distance, etc.

Example Data: WDI, 2019

The [World Development Indicators](#)

- Cross-national country-level data
- $N \approx 215$ countries, one year (2019) + missingness
- Selected variables:
 - Country Name + ISO
 - Region
 - Land Area + Arable Land
 - Population + Pop Growth
 - Birth + Fertility Rates
 - Life Expectancy
 - GDP (Total, Per capita, Growth)
 - Trade + Imports + Exports
 - Expenditures: Military, Education, Health
 - Political Stability + Rule of Law + Control of Corruption
 - Discrete versions of several of the above
- For details, see the [data description](#) in the Github repository

No-Code Visualization Tools

Some alternatives:

1. DataWrapper
 2. Flourish
 3. RawGraphs
 4. Tableau
 5. Power BI (Microsoft)
 6. **Google Sheets**
- ... many, many others

Also: **LLMs**

Create better charts, maps, and tables with ease.

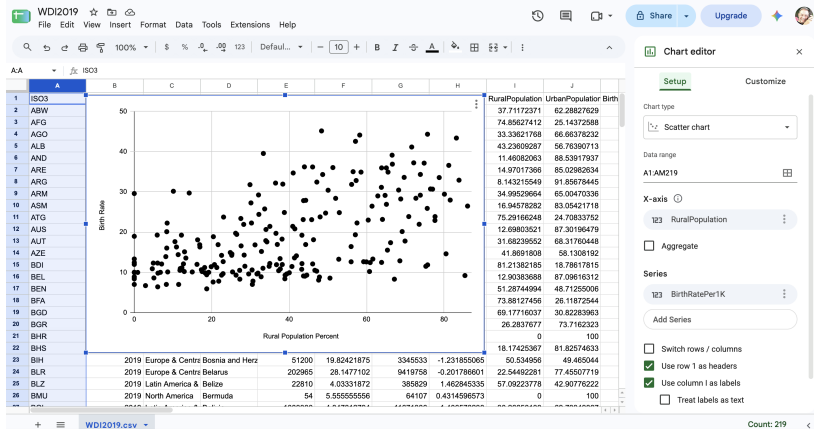
Design and publish high-quality data visualizations with a single, powerful web app.

Start creating →

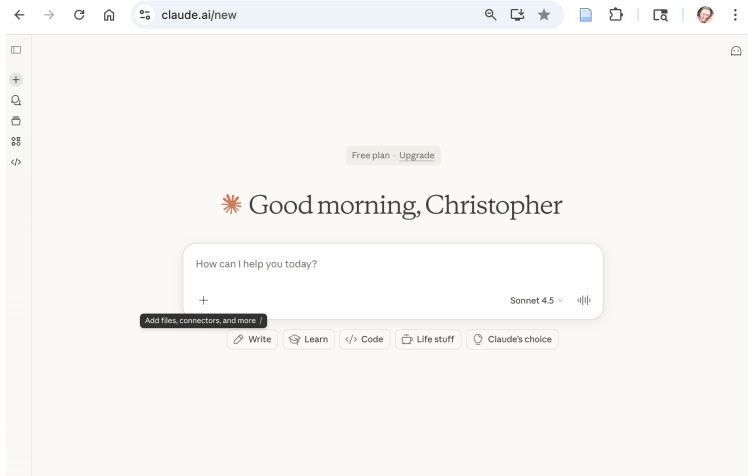
You can try it, no account needed.



Google Sheets (scatterplot)



AI / LLMs (e.g., Claude)



Simple → Complex

Univariate:

- Dotcharts / Barcharts
- Histograms / Density Plots
- Boxplots
- Pie charts / “Donut” charts
- Others... (time series, etc.)

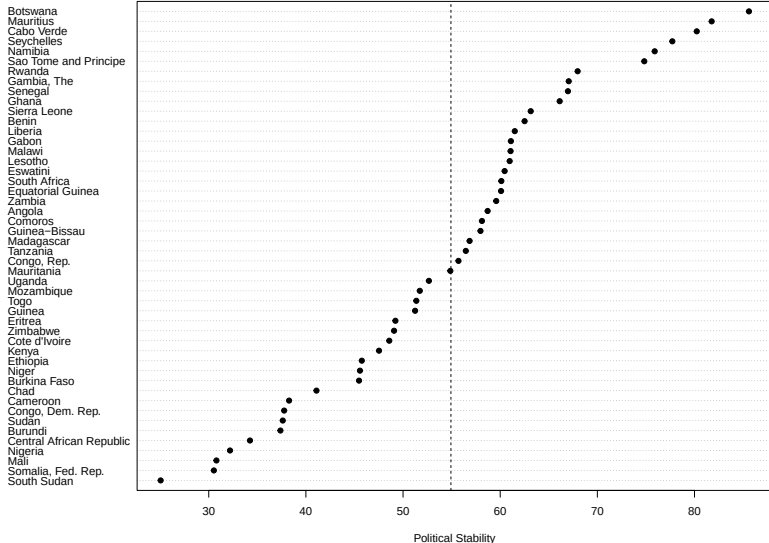
→ Bivariate:

- Range / Arrow Plots
- Scatterplots / Contour Plots
- Conditioned Plots
- Others...

→ Multivariate...

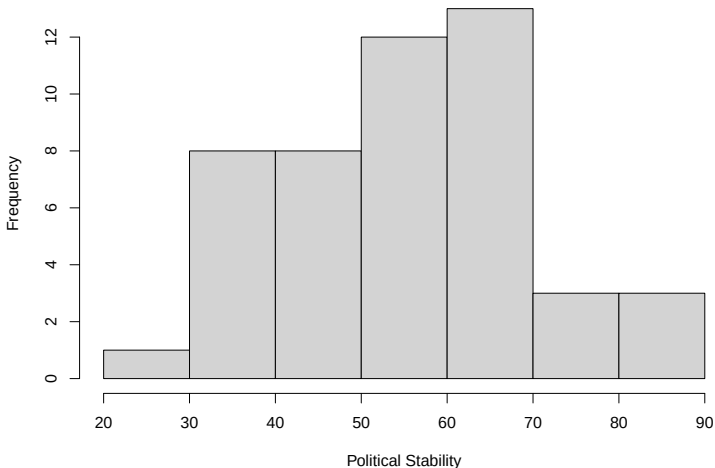
“Raw” Data: Dotplot

Political Stability, Sub-Saharan Africa



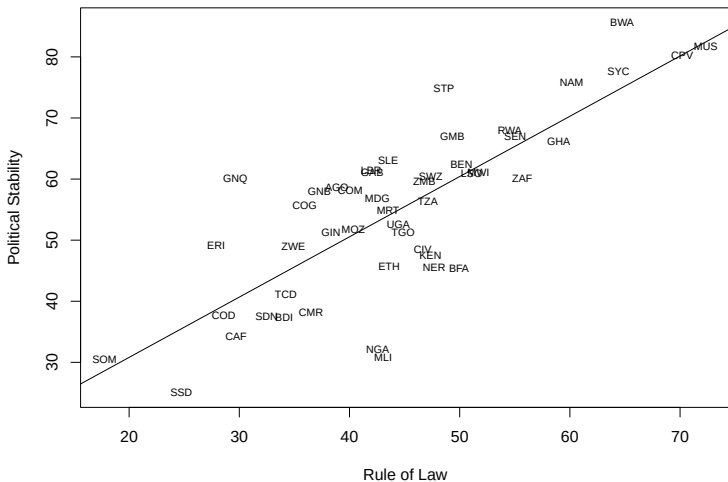
Univariate Summary: Histogram

Political Stability, Sub-Saharan Africa



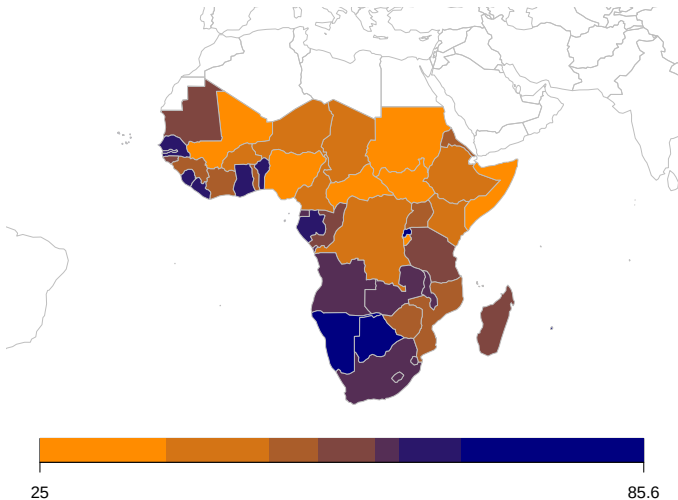
Bivariate Plot: Scatterplot

Political Stability vs. Rule of Law, Sub-Saharan Africa



Geodata: Choropleth

Political Stability, Sub-Saharan Africa (2019)



Good Visualization Practices

Some things to remember:

- Go from simple to complex
- Optimize {readability, information density}
- Each plot should “stand on its own”
- Consider colorblindness (see, e.g., [here](#))